Auteur: Thijs Van Schel
Studierichting: Informatica
Oefeningenreeks 5F nr. 1.9

$$r = 4 + 3\cos\theta \Rightarrow S = \frac{1}{2}\int_0^{2\pi}(4+3\cos\theta)^2$$

$$= \frac{1}{2}\int_0^{2\pi} 16 d\theta + \frac{1}{2}\int_0^{2\pi} 24\cos\theta d\theta + \frac{1}{2}\int_0^{2\pi} 9\cos^2\theta d\theta$$

$$= 16\pi + 12\left[\sin\theta\right]_0^{2\pi} + \frac{9}{2}\int_0^{2\pi}\frac{1+\cos(2\theta)}{2}d\theta$$

$$= 16\pi + 0 + \frac{9}{4}\int_0^{2\pi} d\theta + \frac{9}{4}\int_0^{2\pi}\cos(2\theta)d\theta$$

$$= 16\pi + \frac{18\pi}{4} + \frac{9}{4}\left[\sin(2\theta)\right]_0^{2\pi}$$

$$= 16\pi + \frac{18\pi}{4} = \frac{64+18}{4}\pi = \frac{41}{2}\pi$$

Tekening:

$\theta = 0 \Rightarrow r = 7$

$\theta = \frac{\pi}{2} \Rightarrow r = 4$

$\theta = \pi \Rightarrow r = 1$

$\theta = \frac{3\pi}{2} \Rightarrow r = 4$

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References